Electronic Instrumentation and optoelectronics LAB

LAB1: knowing your LAB workbench

The aim of this LAB session is to introduce your work site. You will analyze the instruments main features and limitations.

The Pre-Lab section is devoted to the analysis of the instruments characteristics, taking always in mind that many signals with we will work are very low amplitude - low frequency signals, in presence of noise and interferences.

PRE-LAB Exercises.

PRELAB1-1.-Describe briefly the function of each of the following instruments: oscilloscope, digital multimeter, function generator, power supply, and data acquisition board. Explain their main characteristics and draw a model of their input or output circuits.

PRELAB1-2.-Look for the following definitions in the international vocabulary of metrology that you have in the readings folder of session 1: accuracy, repeatability, sensitivity, selectivity, and measurement uncertainty.

PRELAB1-3.- Let's suppose you want measure the amplitude a low voltage, low frequency signal (10 mVpp, 1 Hz) you could choose between the oscilloscope and the digital multimeter of your work bench. Consult the specifications of both instruments and try to calculate the measurement uncertainty when the oscilloscope is used and when the digital multimeter is used. Explain which instrument parameters you have considered and detail all your calculations.

PRELAB1-4. - You want to characterize the uncertainty of the function generator. From the instrument specifications try to calculate the uncertainty in amplitude and frequency at the following frequencies 1 Hz, 10 Hz, and 1 kHz. Suppose you have 1 Vrms of amplitude.

PRELAB1-5.- If you measure the previous signal, calculate which is the measurement uncertainty due to the digital multimeter.

LAB exercises

LAB1-1.- Connect the function generator, and select a 2 Vpp, 10 Hz sinus signal. Connect the output of the function generator to the oscilloscope and set the controls to see 1 signal cycle. You must see the signal amplitude as big as possible in the oscilloscope screen. The oscilloscope must be coupled in DC and the trigger must be in the AUTO position. **Do not use the Auto-scale button!!**

Draw the obtained waveform, and write down the control's position.

Vertical amplifier: 250m V/div

Timebase: 13m s/div

DC Coupling

LAB1-2.- Change the oscilloscope coupling to **AC**.
Draw the obtained waveform, and write down the control's position.

Vertical amplifier: 250m V/div

Timebase: 13m s/div

AC coupling

Is there any difference between the two waveforms?
What is the reason?

No, is the same waveform, because in AC and DC the signal has the same waveform

LAB1-3.- Now, change the waveform to an square wave and maintain the oscilloscope settings.

Draw the obtained waveform, and write down the control's position.



Vertical amplifier: 250m V/div

Timebase: 13m s/div

AC coupling

LAB1-4.- Change the oscilloscope coupling to **DC**.
Draw the obtained waveform, and write down the control's position.



Vertical amplifier: 250m V/div

Timebase: 13m s/div

DC Coupling

Explain the reason of the difference between the two measured waveforms.

The reason between the two measured wafeforms it is due to the period of the AC current, and it can be observed the fall of te flat part because of the discharge in the changes of the intermintent current.
És degut a un pas alt, i degut a la càrrega d'un condensador.
At 10Hz you see a change in the frequence.
Tenim el canvi per l'escaló del cas de pujadda i baixada.
Tao=2*pi*RC

A frequencies molt altes disposem d'un low pass filter

LAB1-5.- Measure the waveform rise-time. Compare the obtained value whit that given in the instrument specifications. Are there similar? If not, explain which other factors contribute to the measured rise-time.



LAB1-6.- Connect the power supply and set it to obtain an output of ± 7V. Measure these voltages with the Digital multimeter. Draw the measurement set-up and explain how the measurement has been performed.



LAB1-7.- Set the waveform generator to obtain a square wave with an amplitude of 3 Vpp and a mean value of 1.5 V. Measure this wave with the oscilloscope to verify it. Explain clearly the oscilloscope and waveform generator settings.



LAB1-8.- Connect the multimeter to the waveform generator and measure the root mean square value of the voltage. Explain how the measurement has been performed.

LAB1-9.- Calculate the root mean square value of the previous waveform. Coincides with the measured one? If not, explain why.